

A
Project Report
on

**Mentexa: Arduino-Driven Interactive Embedded System
for Numerical Puzzle Games**

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
in
EEE



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CERTIFICATE

I hereby submit the Project entitled **Mentexa: Arduino-Driven Interactive Embedded System for Numerical Puzzle Games** in the **School of Automation** of the Banasthali Vidyapith, under the supervision of **Dr. Mohit Jain**, School of Automation, Banasthali Vidyapith, Rajasthan, India.

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ABSTRACT

This project introduces an interactive embedded system that modernizes the experience of traditional numerical puzzle games by offering a dynamic, user-friendly digital platform. Designed to enhance cognitive engagement, the system allows users to solve logic-based puzzles in a more intuitive, guided, and responsive environment. It replaces the limitations of static, paper-based gameplay with a responsive setup that provides immediate feedback.

The input from the user is taken with the help of tactile push buttons and is displayed with the help of different coloured LEDs. The colour of the glowing LED depends on the number of times the push button is pressed. The input is then matched with the pattern provided and the result is displayed making it more interactive.

The game encourages users to develop essential cognitive skills such as logical reasoning, pattern identification, strategic thinking, and decision-making. Through interactive gameplay, users receive real-time responses that help identify mistakes and reinforce learning. This system is well-suited for learners, educators, and puzzle enthusiasts, providing a fun yet purposeful tool to stimulate mental development. It can be used in classrooms, learning centers, or even at home as a recreational activity with educational value.

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INDEX

Sr. No.	Contents	Page No.
1.	CERTIFICATE	i
2.	ABSTRACT	ii
3.	ACKNOWLEDGEMENTS	iii
4.	CONTENTS	iv
5.	LIST OF FIGURES	v
6.	CHAPTER I: INTRODUCTION TO THE PROJECT	
1.1	Usage of Arduino in games	
1.2	Objectives and Purpose of the Game	
1.3	Components and Materials	
1.3.1	Hardware Components	
1.3.2	Software Tools	
7.	CHAPTER II: DEVELOPMENT A JOURNEY	
2.1	Circuit Design and Implementation	
2.1.1	Schematic Design and Simulation	
2.1.2	PCB Design and Fabrication	
2.1.3	System Workflow	
2.2	Code Structure and Game Logic	
2.2.1	Game Flow and Logic	
2.2.2	Pre-Hardware Testing Setup	
2.3	Preparation for Hardware Testing	
8.	CHAPTER III: RESULTS AND TESTING	
3.1	Functional Testing	
3.1.1	Hardware Design Performance	
3.1.2	Software-Hardware Integration Testing	
9.	CHAPTER IV: CHALLENGES FACED AND CONCLUSION	
10.	CHAPTER V: REFERENCES	

LIST OF FIGURES

Fig. No.	DESCRIPTION	PAGE NO.
Fig 1.1:	Arduino	2
Fig 1.2(a):	Arduino Nano	
Fig 1.2(b):	74HC595 Shift Register Module	
Fig 1.2(c):	LEDs	
Fig 1.2(d):	Resistors	
Fig 1.2(e):	Tactile Push Buttons	
Fig 1.2(f):	TFT Screen	
Fig 1.2(g):	Relimate connectors	
Fig 1.2(h):	MCP module	
Fig 1.2(i):	7 Segmented LED display (TM1637)	
Fig 1.2(j):	Buzzer	
Fig 1.2(k):	Copper Clad Sheet	
Fig 1.2(l):	Jumper Wires	
Fig 1.2(m):	Zero PCB Sheet	
Fig 1.2(n):	Soldering Setup	
Fig 1.2(o):	Ferric Chloride Chemical	
Fig 1.2(p):	PCB Mate Machine	
Fig 1.2(q):	Multimeter	
Fig 2.1:	Platform for PCB designing	

INTRODUCTION

1. INTRODUCTION

In this project, we have developed Mentexa, an electronic version of the classic Sudoku puzzle game using an Arduino-based system. It is not simply a replica of the traditional paper-based Sudoku but an interactive and enhanced electronic version of it. Sudoku is a logic-based number puzzle that challenges players to fill a grid with digits so that each row, column, and sub-grid contains unique numbers without repetition. Our system recreates this puzzle electronically by incorporating an Arduino microcontroller, push buttons, LED indicators, and various supporting electronic components. The Arduino is responsible for processing player inputs and verifying whether the entered number follows Sudoku rules, while the LEDs or display modules provide feedback to guide the player. This combination of hardware and software ensures an engaging and interactive gameplay experience.

A key aspect of our design is the use of printed circuit boards (PCBs) to organize and connect the electronic components efficiently. A PCB is a fundamental component in electronic circuits, providing a structured and reliable means of connecting components through conductive pathways etched onto a substrate. PCBs offer several advantages over traditional breadboard-based circuits, including durability, compactness, and reduced risk of loose connections. The fabrication process for our PCB involved etching, drilling, and soldering, all of which were performed manually. These hands-on steps enhanced our practical skills in PCB design and assembly, strengthening our understanding of electronics and circuit manufacturing.

Through this project, we gained valuable experience in hardware integration, microcontroller programming, and PCB fabrication. The combination of theoretical knowledge and practical application provided us with a deeper understanding of embedded systems, circuit design, and the challenges involved in real-world electronic development. This project not only demonstrates an innovative electronic adaptation of the Sudoku puzzle but also highlights our ability to design and implement a functional electronic system from scratch.

1.1.USAGE OF ARDUINO IN GAMES

1. Arduino in General Applications

Arduino is an open-source microcontroller platform that has gained widespread popularity due to its ease of use, affordability, and flexibility. It is commonly used in various fields, including:

- Embedded Systems & Automation: Arduino is widely used in home automation, industrial automation, and IoT (Internet of Things) applications to control sensors, motors, and actuators.
- Robotics: It serves as the core of many robotic systems, handling motor control, sensor integration, and decision-making algorithms.
- Prototyping & Product Development: Engineers and hobbyists use Arduino for rapid prototyping to test concepts before developing full-scale electronic products.
- Medical & Environmental Monitoring: Arduino-based systems are used in temperature sensing, heart rate monitoring, and pollution detection.
- Education & Learning: Arduino is an essential tool in learning programming, electronics, and microcontroller-based system design in schools and colleges.

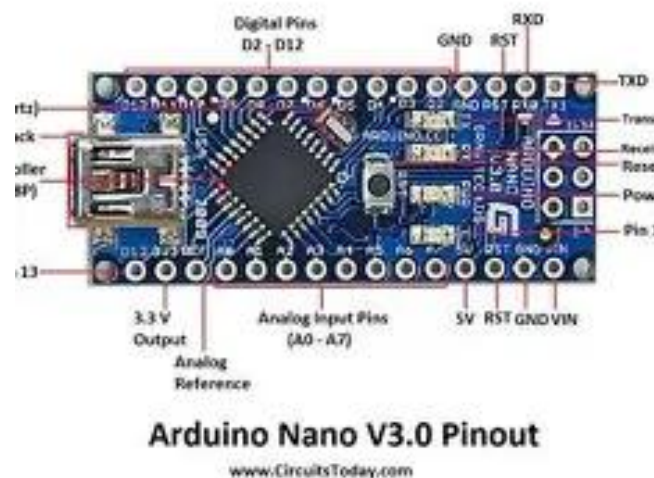


Fig1.1: Arduino

2. Arduino in Electronic Games

Arduino is also widely used in interactive electronic games, allowing developers to create custom hardware-based gaming systems. Some common applications include:

- LED-based Reaction Games: Games that test a player's reaction time by lighting up LEDs and requiring a quick response.
- DIY Arcade Games: Arduino is often used to recreate classic arcade-style games using buttons, joysticks, and LED displays.
- Puzzle & Logic Games: Arduino can be used to build games such as Tic-Tac-Toe, Simon Says, and digital escape room puzzles.
- Gesture & Motion-based Games: By integrating sensors like accelerometers and gyroscopes, Arduino can enable motion-controlled gaming experiences.

Arduino's versatility allows it to serve as the game controller, logic processor, and display interface, making it an ideal choice for custom-built electronic games.

3. Arduino in Our Mentexa Game

In our project, Arduino plays a crucial role in implementing the Mentexa Sudoku game by handling both hardware control and puzzle logic processing:

- Controlling Displays: The Arduino controls the TFT display to show the Sudoku grid and game interface, while the 7-segment display is used to display information such as timer.
- Processing Player Inputs: Player inputs are taken through a 6×6 matrix keypad made of push buttons. The Arduino reads these inputs using the matrix scanning method, allowing the player to select grid positions and enter numbers.
- Using Expansion Modules: To handle the large number of inputs and outputs efficiently, the system uses a 74HC595 shift register module and an MCP23X17 I/O expander module, which allow the Arduino to control multiple LEDs and read additional inputs without using too many microcontroller pins.
- Providing Audio-Visual Feedback: LEDs and a buzzer are used to give immediate feedback to the player. LEDs indicate actions or game status, while the buzzer produces sound alerts for correct entries, wrong moves, or game completion.

- Game Logic Execution: The Arduino processes the Sudoku rules and checks whether the entered number is valid according to the row, column, and sub-grid conditions.
- Ensuring Real-time Response: The Arduino processes inputs and updates the game status instantly, ensuring a seamless player experience.

By integrating the Arduino with TFT display, 7-segment display, matrix keypad, LEDs, buzzer, shift registers, and I/O expander, we have developed a fully interactive electronic Sudoku system. This project demonstrates the practical application of Arduino in building complex embedded gaming systems with multiple hardware interfaces.

1.2. HARDWARE COMPONENTS

1. Electronic Components

- Arduino Nano

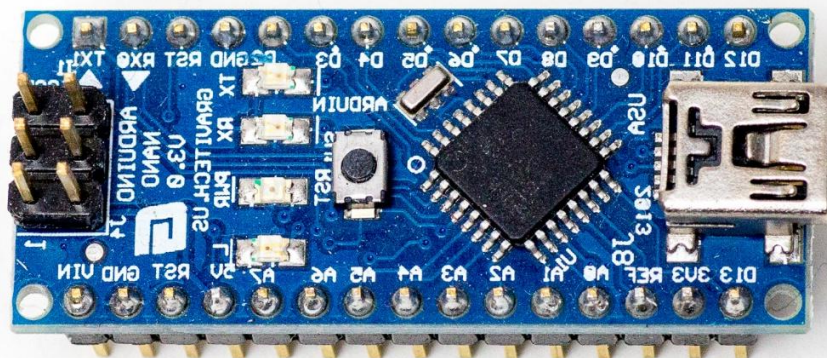


Fig. 1.2(a): Arduino Nano

- 74HC595 Shift Register Module – Expands the number of output pins to control multiple LEDs

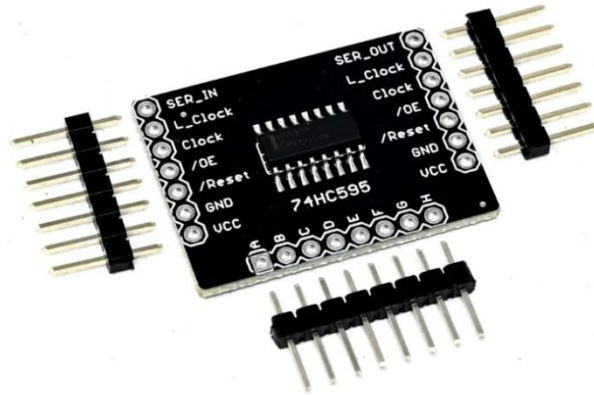


Fig. 1.2(b): 74HC595 Shift Register Module

- LEDs (x6) – Used for displaying input and the number entered.



Fig. 1.2(c): LED

- Current Limiting Resistors (220Ω or 330Ω) – Protects LEDs from excessive current



Fig. 1.2(d): Resistors

- Push Buttons(x36) – Allows user to give input



Fig. 1.2(e): Tactile Push-Button (6x6x5mm)

- TFT LCD Screen (ST7735)



Fig. 1.2(f): TFT LCD screen (ST7735)

- Relimate Connectors

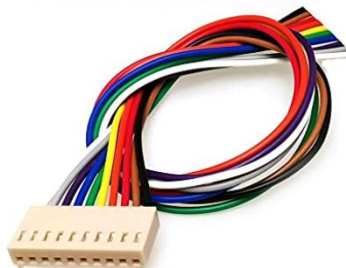


Fig. 1.2(g): Relimate connector

- MCP 23X17

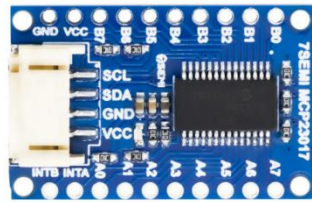


Fig. 1.2(h): MCP Module

- 7-Segmented (TM1637)



Fig. 1.2(i): 7-Segmented (TM1637)

- Buzzer

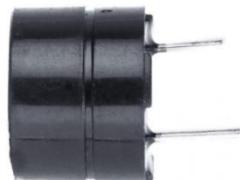


Fig. 1.2(j): Buzzer

2. PCB & Circuit Components:

- Single-sided PCB/Copper Clad Sheet – Used to mount and connect component.



Fig. 1.2(k): Copper Clad sheet

- Jumper Wires (including male-to-male, female-to-female, male-to-female) – For linking the three PCBs together

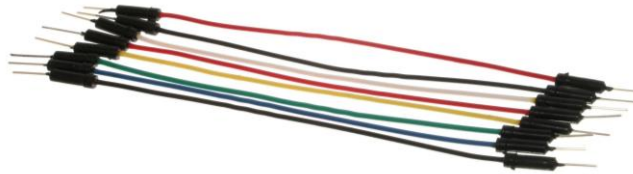


Fig. 1.2(l): Jumper wires

- Zero PCBs (x2) – for

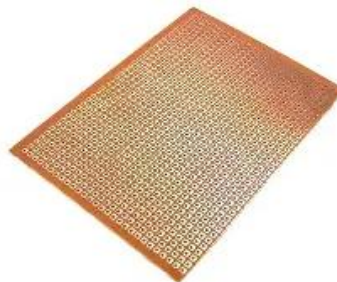


Fig. 1.2(m): Zero PCB sheet

- Soldering Components – Solder wire, flux, and soldering iron for assembling the circuit



Fig. 1.2(n): Soldering Set-up

4. Tools & Fabrication Materials

- PCB Etching Material –In this case, Ferric chloride solution is used.



Fig. 1.2(o): Ferric chloride Chemical

- Drilling Tools – PCB drill for making holes for components

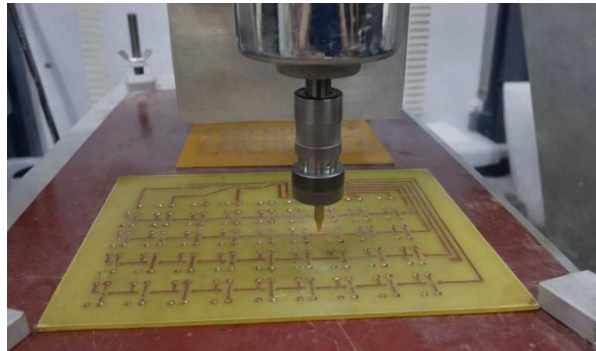


Fig. 1.2(p): PCB Mate Machine

- Multimeter – For testing and debugging connections



Fig. 1.2(q): Multimeter

Chapter II

PCB FABRICATION AND MANUFACTURING TECHNIQUES

Printed Circuit Board (PCB) fabrication is a fundamental process in modern electronics, serving as the backbone for assembling and connecting electronic components in a structured and reliable manner. It provides a compact, robust platform that replaces complex wiring and significantly improves the reliability, manufacturability, and performance of electronic circuits. The PCB fabrication process comprises several stages—starting from circuit design and layout to the final assembly and testing.

1. PCB Design and Layout

The PCB fabrication process begins with the design and layout of the circuit, typically carried out using specialized computer-aided design (CAD) software such as Eagle, KiCad, or Altium Designer. The schematic circuit diagram is first converted into a PCB layout, which defines the physical arrangement and interconnection of components on the board.

The key elements of a PCB layout include:

- *Copper traces*: These are conductive pathways that electrically connect components.
- *Pads and vias*: Pads serve as points for soldering component leads, while vias allow electrical connections between different layers in multi-layer boards.
- *Silkscreen layer*: This is a printed layer on the surface that displays component labels, orientation indicators, and reference designators to assist during assembly.

Upon finalization of the layout, the design is exported as a set of Gerber files, a standard file format used in PCB manufacturing. These files contain precise information about each layer of the board, including copper, solder mask, silkscreen, and drill data.



Fig2.1: platform FOR PCB designing

2. PCB Manufacturing Techniques

Several methods are used to fabricate PCBs, depending on the complexity and resources available.

(a) Etching Method (Chemical Etching) –

It is the most common method used for PCBs for less complex designs as it is not very expensive. We have used this method for fabrication of our project PCBs.

Process:

1. A copper-clad board (a board with a thin copper layer) is cleaned thoroughly so it is free of any scratches.
2. The circuit pattern is transferred onto the board using an iron (a local iron used to iron clothes).
3. Once the design is imprinted on the sheet, then the traces are marked with a CD marker so when it is submerged in the chemical solution, the traces don't wash away.
4. The board is submerged in an etching solution (such as ferric chloride or ammonium persulfate), which removes the unwanted copper, leaving behind only the required traces.
5. The markings on the traces are removed using acetone, and the board is cleaned.

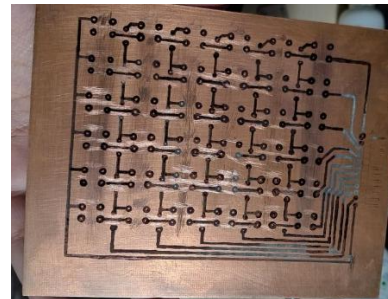


Fig2.2(a) : Tracing the tracks

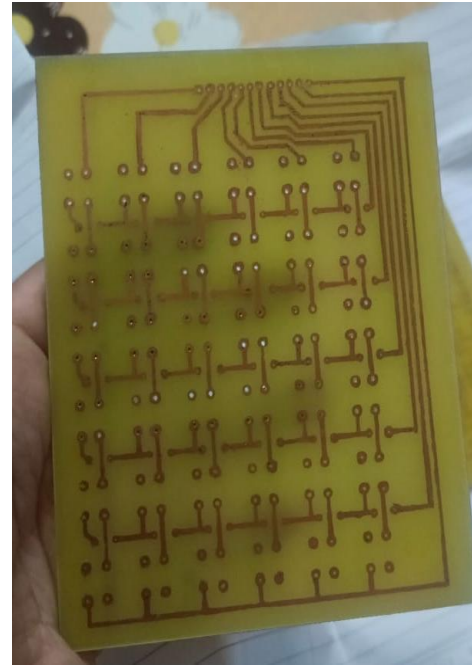
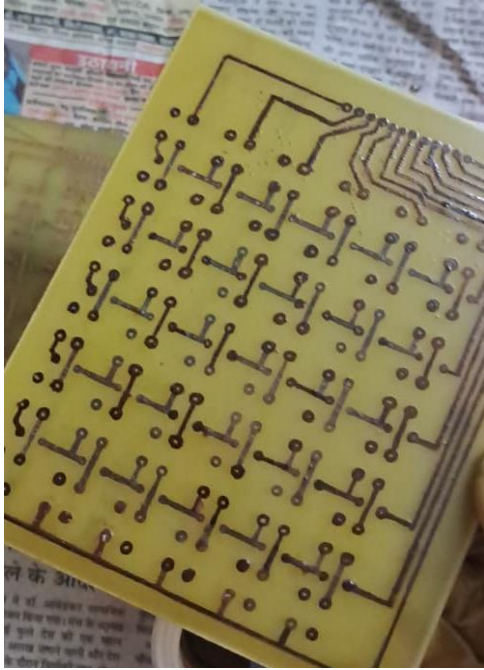


Fig 2.2(b): Tracks after etching process

Advantages:

- It is suitable for small-scale projects.
- It is a Low-cost method.

Disadvantages:

- It is not suitable for complex, multi-layer PCBs.
- It can produce uneven traces if not done carefully.

(b) CNC Milling Method

Process:

1. A CNC milling machine uses fine drill bits to remove unwanted copper from a copper-clad board mechanically.
2. The machine follows a digital tool path derived from the PCB layout to carve out the circuit traces.

Advantages:

- Higher precision than manual etching.
- Eliminates the need for chemical etchants.
- Ideal for rapid prototyping.

Disadvantages:

- Equipment is expensive and requires proper calibration.
- Milling tools wear out over time and may affect consistency.
- Not efficient for high-volume production.

(c) Multi-layer PCB Fabrication – It is generally used in High-End and Industrial Electronics.

Multi-layer PCBs consist of multiple copper layers laminated together with insulating layers in between. These are essential for compact and high-performance electronic systems such as computers, smartphones, communication devices, and automotive systems.

Process:

1. Inner layers are fabricated individually using techniques similar to single-layer PCBs.
2. Layers are stacked and laminated under heat and pressure to form a unified board.
3. Vias (plated-through holes) are drilled and copper-plated to provide interlayer electrical connections.
4. Outer layers undergo etching, solder mask application, and silkscreen printing.

Advantages:

- Supports complex and densely packed circuits.
- Reduces signal interference and enhances performance.
- Enables miniaturization of electronic devices.

Disadvantages:

- Requires specialized fabrication facilities.

- Significantly more expensive than single-layer boards.
- More complex to troubleshoot and repair.

3. Drilling and Component Mounting

Once the circuit traces are prepared, the board is drilled to create holes for mounting components.

Drilling can be done manually using a PCB drill or automated using CNC drilling machines for precise hole placement.

Component mounting methods:

- Through-Hole Technology (THT): Components have long leads that are inserted into drilled holes and soldered on the opposite side of the board. This method provides strong mechanical bonding and is commonly used in educational and industrial projects. In our project we have used this technology.
- Surface Mount Technology (SMT): Components are mounted directly onto the PCB surface. SMT allows for compact designs and is the preferred method for high-volume manufacturing.

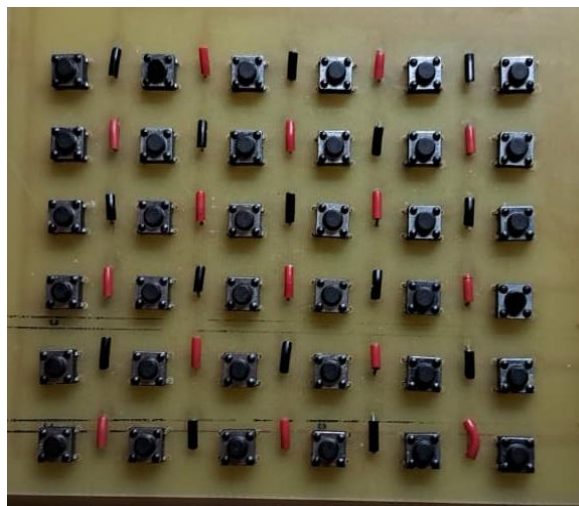


Fig2.3 this is how the components were surface mounted and soldered.

4. Soldering and Assembly

Following component placement, electrical connections are made via soldering:

- Manual Soldering: Performed using a soldering iron, flux and soldering wire. This type of soldering is used in our project.

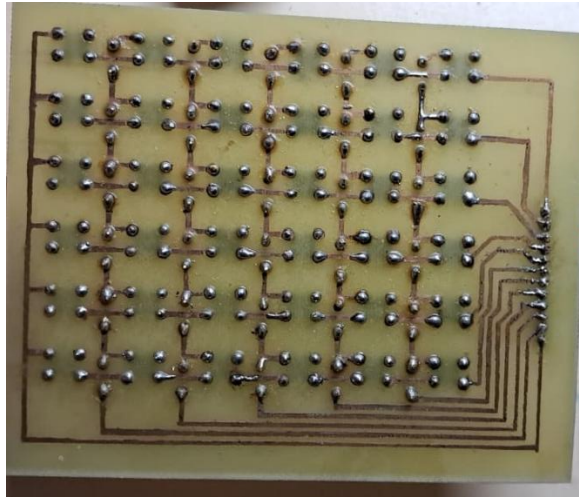


Fig 2.4 Soldered points

- Reflow Soldering: Used in automated environments. Solder paste is applied to pads, components are placed, and the board is passed through a reflow oven where solder melts and solidifies, securing the components.

In some advanced assemblies, wave soldering may also be used for through-hole components.

5. Testing and Quality Control

To ensure the board functions as intended, the following tests are performed:

- Continuity Testing: Verifies that all electrical paths are correctly connected and there are no open circuits.
- Short Circuit Testing: Detects any unintended connections between traces.
- Functional Testing: Involves powering up the circuit and checking its actual operation under expected conditions.

CODE IMPLEMENTATION AND WORKING

3.1 Introduction

This chapter details the implementation of the Mentexa Hardware Sudoku Game using an Arduino-based system and the integration of multiple hardware components including a TFT display, push button matrix, LED indicators via shift registers, a 7-segment display, and a buzzer.

The core logic of the game is implemented in Arduino C/C++, focusing on efficient handling of user input, real-time validation, visual feedback, and time-based gameplay.

The system replaces traditional pen-and-paper Sudoku with an interactive hardware interface, making the gameplay more engaging and intuitive.

3.2 Code Overview

The code is structured around a 6×6 Sudoku logic, where a partially filled grid is displayed and the player must correctly fill the missing values within a limited time.

The code handles:

- Sudoku grid display on TFT
- Input detection using 6×6 push button matrix
- Multi-tap input logic for number selection (1–6)
- LED indication using shift register
- Real-time validation of user input
- Timer management using 7-segment display
- Game state handling (win/lose conditions)

The code is modularized into functions to handle different parts of the game logic, such as:

- `readMatrix()` — detects which push button (cell) is pressed using matrix scanning
- `glowLEDByCount()` — controls LEDs based on number of taps for value selection
- `updateTimer()` — handles the 5-minute countdown timer and display

- drawGrid() — displays Sudoku grid and pre-filled values on TFT
- checkWin() — checks whether the Sudoku is correctly solved
- loop() — manages continuous game execution including input, display, and validation

```
#include <Wire.h>
#include <Adafruit_MCP23X17.h>
#include <Adafruit_GFX.h>
#include <Adafruit_ST7735.h>
#include <SPI.h>
#include <TM1637Display.h>

// ----- TFT -----
#define TFT_CS 10
#define TFT_DC 8
#define TFT_RST 9
Adafruit_ST7735 tft = Adafruit_ST7735(TFT_CS, TFT_DC, TFT_RST);

// ----- MCP -----
Adafruit_MCP23X17 mcp;

#define ROWS 6
#define COLS 6

int rowPins[ROWS] = {0,1,2,3,4,5};
int colPins[COLS] = {8,9,10,11,12,13};

// ----- Shift Register -----
#define DATA 6
#define LATCH 5
```

Fig: Initialization in the code

3.3 Game Algorithm

- **Game Start (Initialization)**

On startup, the system:

Initializes all hardware components

Displays a welcome screen followed by game instructions on TFT

Shows rules such as:

- Use numbers 1–6
- Avoid repetition in rows and columns

- Complete within 5 minutes

After initialization, the Sudoku grid is displayed.

- **Grid Display Phase**

A 6×6 Sudoku grid is drawn on the TFT screen

Some values are pre-filled and displayed in a different color

Remaining cells are left blank for user input

- **User Input Phase**

Each cell corresponds to one push button

The player selects a cell by pressing its button

Input Method:

Repeated presses on the same button determine the value:

- 1 press → 1
- presses → 2
- ...
- presses → 6

After 6 presses, the value cycles back to 1

- **LED Feedback Logic**

6 LEDs are used to indicate selected value:

- 1 press → LED 1 glows
- 2 presses → LED 2 glows
- ...
- 6 presses → LED 6 glows

This provides immediate visual confirmation of the selected number before display.

- **Display and Validation Phase**

Selected value is shown on TFT in the chosen cell

System compares input with correct solution:

- Correct → displayed in Green and stored
 - Incorrect → displayed in Red (not stored permanently)
-

- **Timer Logic**

A 5-minute countdown timer is started at the beginning

Remaining time is displayed on the 7-segment display

If time reaches zero:

- Game ends
 - TFT displays “TIME UP”
-

- **Win Condition**

After each correct entry, system checks completion

If all cells match the solution:

- TFT displays “YOU WIN!”
 - Buzzer produces a success tone
 - Game stops
-

- **Lose Condition**

If time runs out before completion:

- TFT displays “TIME UP!”
 - Game ends without success
-

```
TM1637Display display(CLK_7, DIO_7);

// ----- Buzzer -----
#define BUZZER A1

// ----- Sudoku -----
int sudoku[6][6] = {
    {1,0,0,4,0,6},
    {0,5,6,0,2,0},
    {2,0,1,0,0,4},
    {0,6,0,2,3,0},
    {3,0,2,0,4,0},
    {0,4,5,3,0,0}
};

int solution[6][6] = {
    {1,2,3,4,5,6},
    {4,5,6,1,2,3},
    {2,3,1,5,6,4},
    {5,6,4,2,3,1},
    {3,1,2,6,4,5},
    {6,4,5,3,1,2}
};

int selectedRow=-1, selectedCol=-1, pressValue=0;
```

Fig: Code Snippet

- Integration with Hardware

Component	Function
Push Button Matrix (6×6)	Used for selecting Sudoku cells and input
Shift Register Module (74HC595)	Controls LEDs using fewer pins
LEDs (6)	Indicate selected number based on tap count
TFT Display	Displays grid, inputs, and game instructions
7-Segment Display	Shows countdown timer
Buzzer	Provides audio feedback (win/timeout)
Arduino Nano	Controls logic, input processing, and hardware

```

===== MATRIX READ (FIELD) =====
01 readMatrix(int &outRow, int &outCol){

for(int r=0; r<ROWS; r++){

    // Reset all rows HIGH
    for(int i=0; i<ROWS; i++){
        mcp.digitalWrite(rowPins[i], HIGH);
    }

    // Activate one row
    mcp.digitalWrite(rowPins[r], LOW);

    delayMicroseconds(500);

    for(int c=0; c<COLS; c++){

        if(mcp.digitalRead(colPins[c]) == LOW){

            delay(30); // debounce

            if(mcp.digitalRead(colPins[c]) == LOW){

                outRow = r;
                outCol = c;

                Serial.print("Pressed: Row=");
                Serial.print(r);
                Serial.print(" Col=");
                Serial.println(c);
            }
        }
    }
}

```

Fig: Code Snippet

Working and Flow

1. Startup

- All components are initialized
 - Welcome and rules displayed
 - Sudoku grid loaded
-

2. Input Phase

- Player presses a button corresponding to a cell
 - Repeated presses select value (1–6)
 - LEDs indicate selected value
-

3. Display & Validation

- Value shown on TFT
 - System validates correctness
 - Correct values stored, incorrect rejected
-

4. Timer Execution

- Countdown runs continuously
 - Display updated in real time
-

5. Game End

Win:

- All values correct
- “YOU WIN!” displayed
- Buzzer sound

Lose:

- Time runs out
- “TIME UP!” displayed

• Conclusion

The Mentexa Hardware Sudoku Game demonstrates an effective integration of embedded systems and game logic. The use of a push button matrix for input, LEDs for feedback, and TFT display for visualization creates an interactive and intuitive user experience.

Efficient use of shift registers minimizes hardware complexity, while non-blocking timing ensures smooth gameplay. The system is scalable and can be extended to larger Sudoku grids or additional features, making it suitable for educational and embedded system applications.

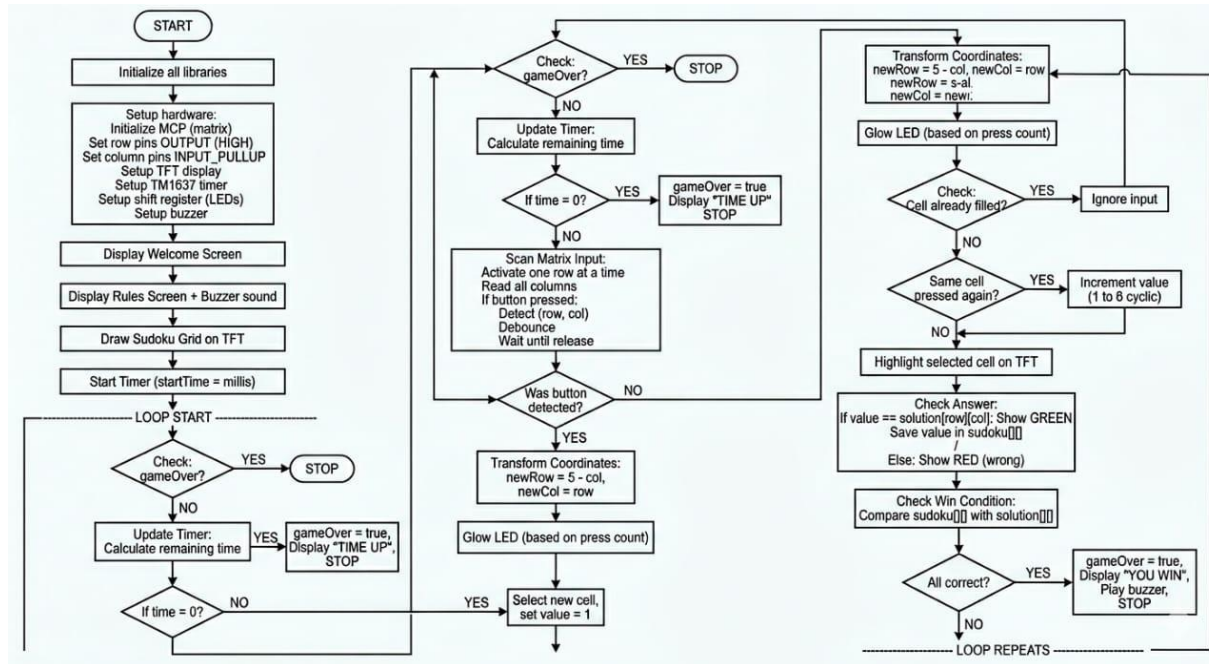


Fig: Algo for the game

HARDWARE TESTING

Hardware testing played a crucial role in validating the functionality, reliability, and responsiveness of each component used in the Mentexa Hardware Sudoku Game. Since the project involves real-time interaction between multiple hardware modules, a systematic and step-by-step testing approach was adopted to ensure proper working and smooth integration of all components.

1. Component-Level Testing

Before integrating the entire system, each hardware component was tested individually to ensure correct functionality:

- Push Button Matrix (6×6)
- Each push button was tested to ensure proper press detection.
- Matrix scanning logic using MCP23017 was verified by checking row-column response.
- Debouncing delays were tested to eliminate false triggers.

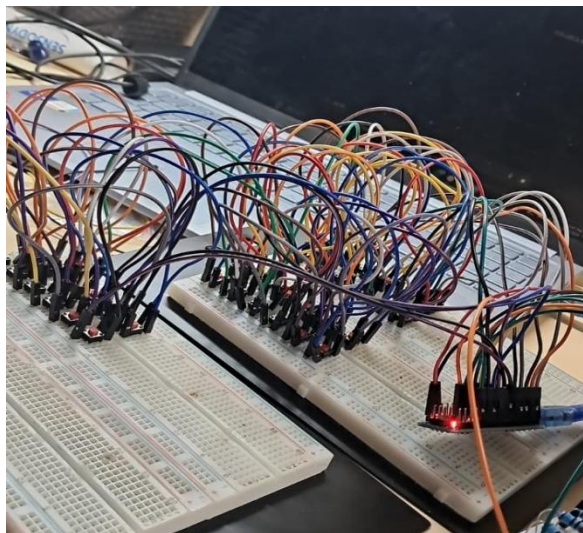


Fig. 6x6 push button matrix testing

- MCP23017 I/O Expander

- The MCP23017 module was tested using I2C communication.
 - Each pin was configured and verified for correct input/output behavior.
 - Row pins were tested as outputs and column pins as inputs with pull-up resistors.
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- LEDs (6 Indicators)

- Each LED was tested individually for brightness and proper functioning.
- LEDs were verified to glow correctly based on tap count (1 to 6).

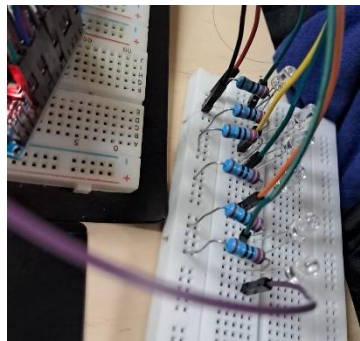


Fig. LEDs testing

- 74HC595 Shift Register Module

- The shift register was tested using basic LED shifting programs.
 - Serial data transmission (DATA, CLOCK, LATCH) was verified.
 - Proper LED control through bit shifting was ensured.
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- TFT Display (SPI)

- The TFT display was tested using graphics libraries.
- Text rendering, grid drawing, and color display were verified.
- SPI connections (MOSI, SCK, CS, DC, RST) were checked for accuracy.

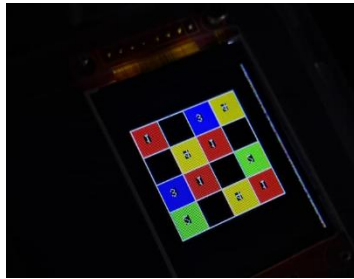


Fig. TFT testing

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- 7-Segment Display (TM1637)
 - The display was tested to ensure correct numerical output.
 - Timer values were verified in MM:SS format.
 - Brightness and segment control were adjusted.

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- Buzzer
 - The buzzer was tested using tone functions.
 - Different frequencies were checked for startup, win, and timeout alerts.

2. Integration Testing

After successful individual testing, components were integrated step-by-step to validate system-level performance:

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- Matrix Input + Display
 - Push button matrix was connected with MCP23017 and integrated with TFT display.
 - Pressed cell positions were verified to map correctly on the Sudoku grid.

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- Input + LED Feedback
 - Button presses were linked with LED outputs via shift register.
 - Number of presses correctly triggered corresponding LED indication (1–6).

- Ensured no incorrect LED activation or lag in response.
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- Display + Game Logic

- TFT display was tested with Sudoku logic:
 - Correct values displayed in green
 - Incorrect values displayed in red
 - Grid updates were verified in real-time.
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- Timer + Display

- Timer was integrated with TM1637 display.
 - Countdown was verified for accuracy and smooth updating.
 - Timeout condition was tested to ensure proper game termination.
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- Full System Testing

- Complete gameplay was tested:
 - Input → Processing → Display → Feedback
 - Verified smooth interaction between all modules without delay or conflict.
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3. PCB Testing and Debugging

After breadboard prototyping, the system was tested on PCB for real-world reliability:

- Initial Dry Run

- Continuity test performed using multimeter.
 - Checked for short circuits and broken connections.
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- Component Soldering Verification

- Each solder joint was inspected carefully.
- Components were tested after incremental soldering to avoid complex debugging later.

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- Power Supply Check
 - Verified stable 5V supply across all components.
 - Ensured no voltage drops or overheating issues.
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- Final Functional Testing
- Complete system was tested with final code.
- Full gameplay executed:
 - Grid display
 - Button input
 - LED feedback
 - Timer countdown
 - Win/lose conditions
- **Minor adjustments were made in:**
 - Debounce delay
 - LED timing
 - Display refresh

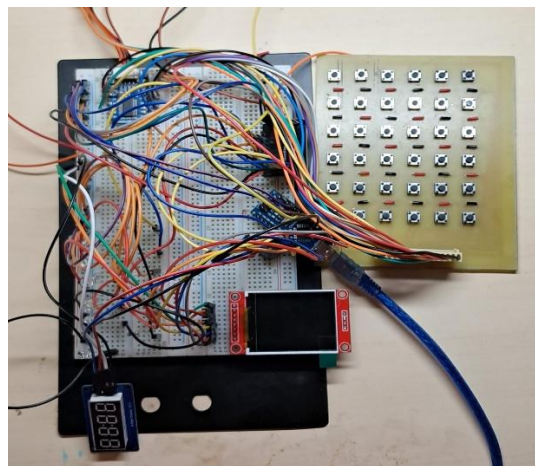


Fig. Final testing



Fig. Final form after casing

Conclusion

Hardware testing was essential in ensuring the stability and reliability of the Mentexa system. By testing each component individually and then integrating them step-by-step, potential issues were identified and resolved early in the development process.

This structured testing approach improved system performance, ensured accurate input handling, and provided smooth real-time feedback. The final prototype operates efficiently and demonstrates a robust combination of embedded hardware and game logic, making it suitable for further enhancements and scalability.

Chapter IV

CHALLENGES FACED AND FUTURE ASPECTS

Challenges Faced During Development

Developing this project was not without its share of difficulties. As a student-driven, hands-on hardware implementation, one of the most significant challenges was understanding and executing PCB (Printed Circuit Board) design and fabrication. This involved not only learning how to lay out circuits using software tools like KiCad, but also ensuring correct routing, pin mapping, and proper component footprint selection. Mistakes such as reversed polarity, incorrect track widths, or improper ground planes had to be identified and iteratively corrected through simulations and prototype testing.

Another key learning curve involved working with shift register module(74HC595) to control multiple LEDs with limited Arduino I/O pins. Efficiently managing the timing and data flow while simultaneously reading button inputs and updating the TFT screen was a delicate task, especially when trying to avoid delays and keep the system responsive using millis()-based timers.

A major challenge in developing Mentexa was designing the 6×6 keypad, as such a module was not readily available online. Therefore, we had to build it from scratch using individual tactile push buttons arranged in a matrix configuration. Designing the PCB tracks for this matrix required careful planning to ensure correct row–column connections so that the Arduino could scan and detect each button accurately without short circuits.

The keypad PCB required drilling and soldering a large number of connections, and in total we had to solder around 216 holes for the keypad network. Completing this soldering with consistent joints while avoiding short circuits demanded patience and precision.

Handling multiple hardware components with limited Arduino I/O pins was also challenging. To address this, we used a 74HC595 shift register and an MCP23X17 I/O expander module, which required careful programming and debugging to ensure smooth communication with the Arduino.

Additionally, integrating the TFT display and 7-segment display with the keypad system required proper synchronization so that the Sudoku grid and game information could update smoothly without affecting input responsiveness.

Despite these challenges, the process provided valuable experience in PCB design, soldering, embedded programming, and hardware debugging, leading to the successful development of Mentexa.

Future Aspects and Scope for Enhancement

Although Mentexa is fully functional in its current form, the project offers strong potential for future improvements and expansion. One possible enhancement is the introduction of multiple difficulty levels. The difficulty can be increased by providing Sudoku boards with more missing digits, which requires deeper logical reasoning and makes the puzzle more challenging for experienced players. Additional gameplay features such as time-based challenges, limited-move modes, or hint systems can further improve engagement and make the game suitable for players of different skill levels.

Mentexa can also be expanded into a multiplayer mode similar to games like Tic-Tac-Toe. In this version, players can take turns solving the puzzle using a pass-and-play method, where the device is passed between players. The player who completes the puzzle in the least amount of time or with the fewest mistakes can be declared the winner, making the gameplay more competitive and interactive.

Future versions of the project can also improve the hardware design. The PCB can be made more compact and optimized, and the keypad design can be refined to make the interface more ergonomic and user-friendly. Additional features such as memory storage for saving puzzle progress, different Sudoku grid sizes, or custom puzzle generation algorithms could also be implemented to make the system more advanced.

From an educational perspective, Mentexa can also be developed as a STEM learning kit for schools, workshops, and maker spaces. Students could assemble the hardware, learn PCB fabrication, understand matrix keypad connections, and practice Arduino programming and embedded system design while interacting with a logic-based puzzle game.

Projects like Mentexa also contribute to the growth of India's technology ecosystem. Developing such locally designed electronic products encourages innovation, practical skill development, and small-scale hardware manufacturing within the country. If projects like this are promoted and commercialized, they could support national initiatives such as Make in India, Digital India, and Atmanirbhar Bharat. By promoting indigenous design, local production, and technological creativity, such innovations can strengthen the Indian electronics sector and help move the country toward a more self-reliant and innovation-driven future.

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